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Effects of objects and image quality on melanoma classification using Spatio Temporal Joint graph Convolutional Network

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ABSTRACT

Malignant melanoma, a highly lethal subtype of skin cancer, is categorized by the abnormal growth of melanocyte cells and is associated with elevated mortality rates. Improving the prognosis of patients with melanoma requires an early and precise diagnosis. Recently, deep learning models have emerged as dominant tools in computer-aided diagnosis (CAD) systems for the categorization of potential melanoma lesions. In clinical settings, obtaining high quality skin images can be challenging due to blurry, noisy, low-contrast images that may contain extraneous objects such as rulers or hair. To overcome these challenges, the Spatio Temporal Joint Graph Convolutional Network (STJGCN) was utilized to detect and classify melanoma from dermoscopy images. The input images are collected from the 2020 Challenge Dataset of the International Skin Imaging Collaboration (ISIC). The input images are provided to Information Exchange Multi-Bernoulli-Filter (IEMBF) based preprocessing for hair noise removal, resizing, random rotations, cropping, and splitting from input dataset. The pre-processed images are provided to Spatio Temporal Joint Graph Convolutional Network (STJGCN) for detecting and classifying melanoma as benign and malignant. The Red Panda Optimization Algorithm (RPOA) is used to fine-tune STJGCN parameters. The proposed EOIQ-MC-STJGCN approach is implemented in Python and various performance metrics are analyzed to assess the efficacy of the proposed approach. The proposed approach attains the highest accuracy of 97.86% in melanoma detection and classification compared to various existing approaches. By demonstrating the efficiency of the proposed EOIQ-MC-STJGCN technique, this research contributes to faster and more accurate melanoma diagnosis, ultimately facilitating earlier intervention and treatment.

1. Introduction

The World Health Organization (WHO) states that skin cancer is a major public health concern because it makes up about 33.33 % of all cancer cases worldwide. The prevalence of skin cancer has significantly increased during the last ten years, particularly in countries like the United States, Australia, and Canada. To mitigate the effects of this escalating health crisis, there is a critical need for heightened awareness, effective prevention strategies, and advanced diagnostic methods, as underscored by rising incidence rates [1,2]. Skin cancers can be broadly categorized into malignant and non-malignant types, with malignant tumors posing a greater health risk. Malignant tumors can spread to another part of body through the lymphatic or circulatory systems after infiltrating nearby tissues. Metastasis is the process by which cancer cells migrate throughout the body and develop into new tumors in other organs. This can lead to serious health problems. To improve patient outcomes due to this aggressive behavior, early detection and treatment are crucial. Squamous cell carcinoma, basal cell carcinoma,

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dermatofibrosarcoma, and malignant melanoma are examples of malignant skin cancers [3]. Malignant melanoma stands out among these because of its propensity for quick growth and metastasis, which, if not identified early, can have a major effect on survival rates. However, benign tumors often never penetrate nearby tissues or spread to other areas of the body, even though they can cause aesthetic difficulties. They grow locally and are generally encapsulated, which makes them easier to remove surgically. While benign tumors can sometimes lead to complications or discomfort, they are not life-threatening and are usually associated with a favorable prognosis. Examples of benign skin tumors include seborrheic keratosis, moles (nevi), and lipomas [4]. However, it is important to note that benign tumors can sometimes be mistaken for malignant ones, so careful examination and, if necessary, a biopsy is essential to differentiate between the two. It is crucial to differentiate between benign and malignant melanoma to provide appropriate therapy and care [5]. The ozone layer's thinning, which permits more ultraviolet (UV) radiation to reach the Earth's surface, is one of the main reasons why skin cancer is becoming more common. UV radiation is a well-known carcinogen that damages skin cells and induces genetic abnormalities, thereby increasing the risk of skin cancer. The correlation between UV exposure and the development of malignant cells highlights the necessity for implementing effective sun protection measures [6]. Various lifestyle and environmental factors exacerbate the harmful effects of UV radiation. Conditions such as infections, alcoholism, smoking, and certain medical disorders can also contribute to the formation of malignant cells. Environmental factors like pollution and exposure to toxins significantly elevate the hazard of skin cancer. To reduce the rising incidence of skin cancer, these modifiable risk factors must be addressed through focused interventions and public health campaigns. The severity of the issue is evident, with an estimated 15,000 deaths attributed to skin cancer annually. While malignant melanoma is less prevalent than other skin cancers, it carries a substantially higher mortality rate [7,8]. Regular skin screenings and awareness campaigns are highly important because early detection is critical to improving prognosis and treatment outcomes. Recent advancements in computer-assisted diagnostic technologies have enhanced the accuracy with which dermatologists can detect skin cancer. These systems utilize advanced algorithms to analyze skin lesions, improving diagnostic impartiality, efficiency, and accuracy by addressing the limitations of traditional clinical evaluations. By categorizing melanoma images into benign and malignant classes, these technologies facilitate more accurate assessments and enable timely interventions, which are vital for improving patient outcomes [9]. To mitigate the impact of this emerging health issue, immediate actions are necessary, including public awareness campaigns, preventive measures, and the development of sophisticated diagnostic tools. International collaboration is imperative to tackle the multifaceted causes of the increasing skin cancer incidence and eventually lessen the burden of this disease on individuals and healthcare systems worldwide [10].

Recent deep-learning approaches for melanoma classification have employed various models and architectures, yet they present several challenges. Convolutional Neural Networks (CNN) and Generative Adversarial Networks (GAN) are extensively used to augment data for classification; however, these methods require high computational power, and GAN produces unrealistic samples that degrade model accuracy. DCNN, utilizing EfficientNet for improved scaling and performance, faces limitations in handling the high variability and class imbalance in skin lesion datasets, often necessitating significant hyperparameter tuning for optimization. Hybrid models, such as those combining ResNet50 and MRCNN for segmentation, add complexity but are computationally intensive and sensitive to dataset quality, requiring extensive labeled data for effective training. Ensemble CNN architectures that combine pre-trained and customized models improve generalization but increase model size, making real-time applications challenging due to memory and processing demands. DNN models trained with transfer learning accelerate training, but they suffer from overfitting if the training dataset is small. Techniques using SVM with residual networks and pre-processing like dilatation to enhance lesion features offer robust feature selection but are hindered by parameter tuning demands in Bayesian optimization, which is computationally costly. Approaches addressing data imbalance and complex feature extraction often necessitate a significant pre-processing pipeline, increasing the overall model complexity and hindering the scalability of diverse datasets. These disadvantages have inspired this work.

In this manuscript, the novelty of the proposed EOIQ-MC-STJGCN approach for melanoma detection lies in the integration of the Red Panda Optimization Algorithm (RPOA) to fine-tune the parameters of the STJGCN. This integration improves classification accuracy by strengthening the model's capacity to extract intricate information from dermoscopy images. The method leverages STJGCN to capture both temporal and spatial information while employing the Information Exchange Multi-Bernoulli Filter (IEMBF) for advanced pre-processing, effectively addressing noise and extraneous objects in clinical images. This unique strategy and integrated model architecture advance melanoma classification, validated against the ISIC 2020 Challenge Dataset.

Major contribution of this manuscript a follows;

- The impact of objects and picture quality on the categorization of melanoma using Spatio Temporal joint Graph Convolutional Network (EOIQ-MC-STJGCN) is proposed.
- ISIC 2020 Challenge Dataset was utilized to ensure that the method is trained and tested on a diverse and challenging dataset for robust melanoma classification.
- Information Exchange Multi-Bernoulli-Filter (IEMBF) was utilized for effective pre-processing, which enhances image quality and removes extraneous objects.
- STJGCN is used for classifying the detected melanoma as the benign and malignant.
- RPOA is integrated to fine-tune STJGCN parameters, which optimizes the model performance, enhancing accuracy and reliability in melanoma detection and classification.

Remaining portion of this work structured as: Section 2: literature survey, Section 3: explains proposed methodology, Section 4: explains results and discussion and Section 5: conclusion.

2. Literature survey

In the literature, several works have been suggested for the deep learning-based classification of melanoma; a few of these are included here,

Saeed et al. [14] have presented utilizing CNNs to accurately detect and categorize different kinds of skin lesions according to the most recent benchmark datasets, ISIC 2019 and 2020. Two advanced Generative Artificial Intelligence (AI) augmentation methods, Enhanced Super-Resolution GAN and GAN, were employed as part of the comprehensive pre-processing of the dataset.

SM et al. [15] have suggested a DCNN-based model that detects and distinguishes between skin cancer types that were melanoma and those that were not. The study's datasets were from the well-known ISIC-2019 and ISIC-2020 challenges, which include problems with unequal class distribution and different image resolutions. The EfficientNet design was employed, which was based on transfer learning techniques, to solve these issues and attain high classification performance. A ranger optimizer was employed to further increase the EfficientNet's efficiency, which significantly lowers the need for hyperparameter adjustment.

Akram et al. [16] have investigated a hybrid deep learning model that blends two modern techniques: For semantic segmentation, use the Mask Region-based Convolutional Neural Network (MRCNN) and ResNet50 for lesion identification. For border delineation, the MRCNN was employed to identify a skin lesion's precise location. For comprehensive model training, a vast annotated collection of dermoscopic

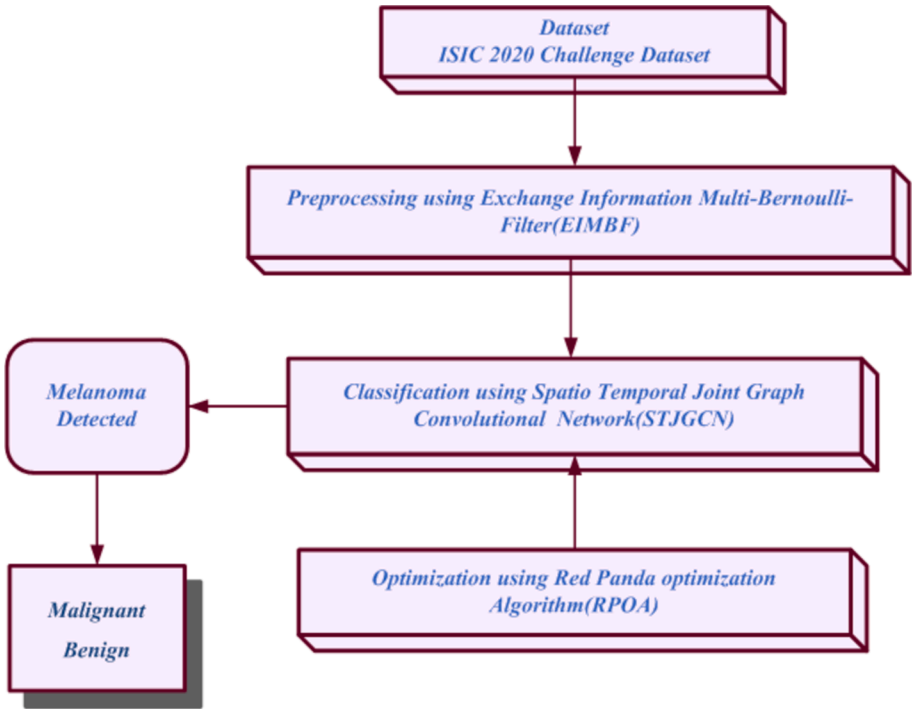


Fig. 1. Block diagram for proposed EOIQ-MC-STJGCN.

images was gathered.

Qureshi and Roos, [17] have suggested a novel ensemble-based CNN design where a meta-learner uses auxiliary data in the form of input picture information to merge many CNN models, a few of which have prior training and others of which were trained solely on the available data. The suggested technique enhances the model's capacity to handle sparse and imbalanced input. The advantages of the recommended method were illustrated using a dataset of 2056 people's 33,126 dermoscopic images.

Venugopal et al. [18] have suggested a deep neural network (DNN) model to diagnose skin cancer using dermoscopic images that have improved learning performance and optimized training. To train the DL models, a knowledge base was created utilizing a range of dermoscopic datasets. On a smaller training dataset, the suggested model was trained more quickly through the employ of transfer learning and fine-tuning. The model performed better when data augmentation approaches were used.

Alenezi et al. [19] have presented Dermoscopic images of skin lesions were utilized in a multi-stage melanoma recognition method. A useful pre-processing method for revealing details and eliminating hair characteristics in dermoscopic pictures was created using dilatation and pooling layers. The corrected images were then subjected to feature extraction using a deep residual neural network. On a smaller training dataset, the suggested model was trained more quickly through the utilize of transfer learning and fine-tuning. The model performed better when data augmentation approaches were used.

Alsahafi et al. [20] have suggested a novel approach to skin lesion classification based on artificial intelligence. A Residual Deep Convolution Neural Network was employed in this technique. As an alternative to moving filters down the horizontal axis, sliding dot product filters to create multiple convolution filters were used for cross-channel correlation and multi-layer feature extraction. The dataset was converted from image and label to image and weight vector, the suggested approach solves the imbalanced dataset issue.

Table 1
Class Distribution of the ISIC 2020 Challenge Dataset.

Class	No of Images
Malignant	584
Benign	32,542
Total	**33,126**

3. Proposed methodology

This section discusses how objects and image quality affect the Spatio Temporal Joint Graph Convolutional Network's ability to classify melanoma. In the proposed methodology section dataset, Filter, Network, and Optimization are described. The inter-sequence analysis approach has several potential applications and can significantly increase efficiency and reduce costs when used for the Classification of Melanoma. The proposed EOIQ-MC-STJGCN classification model aims to increase the classification of Melanoma accuracy and is a useful method of increasing the model's practicability. Fig. 1 displays the Block diagram for EOIQ-MC-STJGCN.

3.1. Image Acquisition

The first step is to collect the ISIC 2020 Challenge Dataset photos that were submitted [21]. Melanoma Institute Australia, Medical University of Vienna, Memorial Sloan Kettering Cancer Center, Hospital Clinic de Barcelona, the University of Queensland, and Athens Medical School were the sources of the dataset, which was produced by the International Skin Imaging Collaboration (ISIC). There are 33,126 dermoscopic training photos in the dataset, representing over 2,000 individuals with different benign and malignant skin lesions. A distinct patient identification linked to each image serves as a means of identifying these people. Histopathology has validated benign diagnoses as well as all expert agreement, malignant diagnoses, and longitudinal follow-up. The Class distribution of the ISIC 2020 Challenge Dataset is displayed in Table 1.

3.2. Pre- processing using Information Exchange Multi- Bernoulli filter (IEMBF)

In this Section, the IEMBF is discussed [22]. The IEMBF is used for hair noise removal, resizing, random rotations, cropping, and splitting from the ISIC 2020 Challenge Dataset. The IEMBF enhances image quality in dermoscopy by filtering out noise, such as hair artifacts, and distinguishing skin lesions from unwanted noise, thus improving clarity for training. IEMBF retains important details during resizing, minimizing distortion and preserving skin lesion features. Integrating IEMBF with data augmentation techniques like random rotations and cropping maintains crucial features while generating diverse training samples. The IEMBF capability to process multiple targets simultaneously allows for efficient handling of large datasets, significantly reducing computational time during transformations like resizing and splitting images. The ISIC 2020 dermoscopy images were imported and ensured that the associated labels for each image were correctly formatted and accessible. The parameters for the IEMBF were defined, including Noise characteristics, detection thresholds, and Initial state estimates for skin lesions. Bernoulli filters for each skin lesion were created in the images. It is given in Eq. (1).

$$\widetilde{e}_{j|j-1}\left(\widetilde{W}_j\right) = \prod_{h=1}^{m_{j|j-1}} \widetilde{e}^{h}_{j|j-1}\left(\widetilde{W}^{h}_j\right) \tag{1}$$

here, $e_{j|j-1}(\cdot)$ represent the multi Bernoulli filter and $\widetilde{W}_j$ denotes the threshold variables, and h is the Noise characteristics. The IEMBF is applied to process each image to identify and filter out hair artifacts. It is given in Eq. (2).

$$\widetilde{e}^{h}_{j|j-1}\left(\widetilde{W}_j\right) = \begin{cases} \left\{1 - q^{h}_{j|j-1} \right. & \widetilde{W}_j = \varphi \\ q^{h}_{j|j-1} o^{h}_{j|j-1}(x)\delta_h[t] & \widetilde{W}_j = \{(t,w)\} \\ 0 & otherwise \end{cases} \tag{2}$$

here, $\delta_h[t]$ represent the hair artifacts, $e_{j|j-1}(\cdot)$ represent the multi Bernoulli filter, $\widetilde{W}_j$ denotes the threshold variables, h is the Noise characteristics, q is the Bernoulli filter and t is time variables. The IEMBF standardize all images to a consistent size suitable for input into models. It is given in Eq. (3).

$$\int \widetilde{e}_{j|j-1=1}\left(\widetilde{W}\right)\delta\widetilde{W} = \sum_{n=o}^{\infty}\frac{1}{n!}\int \widetilde{e}_{j|j=1}(\{w_1, \ldots\ldots, w_m\})cw_{1:m} = 1 \tag{3}$$

here, $e_{j|j-1}(\cdot)$ represent the multi Bernoulli filter, $cw_{1:m}$ is the resized image, t is the time, and $\widetilde{W}_j$ denotes the threshold variables. The random rotation, cropping and splitting is applied to create variations of the cleaned images, different views of the lesions while preserving critical information and Split images into smaller patches, allowing localized analysis of lesions. The IEMBF ensure that key features remain intact during rotation; cropped areas still capture relevant portions of the skin lesions and maintain important features in each patch. It is given in Eq. (4).

$$\widetilde{e}_{j|j}\left(\widetilde{W}_j\right) \alpha \prod_{i=1}^{L} k\left(y^{i}_{j}|\sum_{(t,w)\in\widetilde{W}_j} g^{i}(w)\right) \prod_{h=1}^{mj|j-1} \widetilde{e}^{h}_{j|j-1}\left(\widetilde{W}^{h}_j\right) \tag{4}$$

here,$\widetilde{W}_j$ denotes the threshold variables, and $e_{j|j-1}(\cdot)$ represent the multi Bernoulli filter, h is the Noise characteristics, and t is the time, and L is the augmented image. Information exchange between multiple Bernoulli filters is utilized to enhance state estimates across augmented versions of images. The estimates are continuously updated based on new measurements from augmented image, refining understanding of each lesion. It is given in Eq. (5).

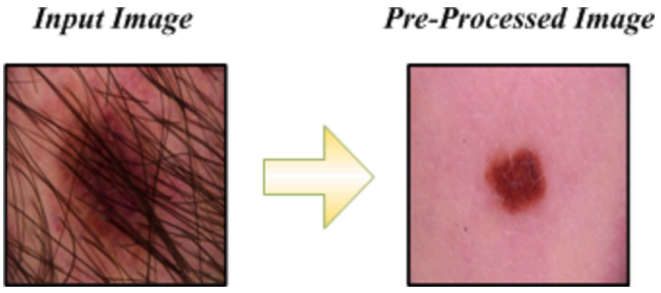


Fig. 2. Input and Preprocessed Images of ISIC 2020 Challenge dataset.

$$\widetilde{e}^{t}_{j|j}\left(\widetilde{W}^{t}_j\right) = \frac{o^{t}\left(y_j|\widetilde{W}^{t}_j\right)\widetilde{e}^{t}_{j|j-1}\left(\widetilde{W}^{t}_j\right)}{\int \widetilde{o}\left(y_j|\widetilde{W}^{t}_j\right)\widetilde{e}^{t}_{j|j-1}\left(\widetilde{W}^{t}_j\right)\delta\widetilde{W}^{t}_j} \tag{5}$$

here,$e_{j|j-1}(\cdot)$ represent the multi Bernoulli filter, $\widetilde{W}_j$ denotes the threshold variables, o^t is the state estimate, e is the updated estimate,t is the time and $\widetilde{o}$ is the original image. Original and augmented images are combined into a single dataset. The output pre-processed is obtained using Eq. (6).

$$C\left(e_{j|j}|p\right) \leq C\left(\widetilde{e}_{j|j}|\widetilde{p}\right) \tag{6}$$

here, p is the input image, $e_{j|j-1}(\cdot)$ represent the multi Bernoulli filter, $\widetilde{q}$ Bernoulli form, and C is the pre-processed image. The Input and Pre-processed Images of ISIC 2020 Challenge dataset is shown in Fig. 2.

The IEMBF has significantly enhanced the preprocessing of dermoscopy images by hair noise removal, resizing, random rotations, cropping, and splitting from the ISIC 2020 Challenge Dataset, leading to enhanced reliability and accuracy in melanoma classification. The pre-processed images are presented to STJGCN for melanoma detection and classification.

3.3. Classification using Spatio Temporal Joint Graph Convolutional Network (STJGCN)

In this section, melanoma detection and classification using STJGCN [23] is discussed. The STJGCN detects melanoma from the pre-processed images of the ISIC 2020 Challenge Dataset and classify such as Benign and Malignant. STJGCN effectively captures spatial and temporal characteristics from skin lesion images, enhancing the understanding of lesion characteristics essential for distinguishing between malignant and benign cases. By modeling relationships among lesions through graph structures, STJGCN achieves higher classification accuracy than traditional CNN. STJGCN is highly scalable, making it suitable for large datasets like ISIC 2020 and adaptable as more data becomes available. The STJGCN architecture starts with the development of both adaptive spatiotemporal joint graphs (STJG) and pre-defined ones that depict dynamic node interactions. Learning dependencies across a range of temporal ranges is made possible by the dilated causal spatiotemporal joint graph convolution layers. A multi-range attention mechanism combines data from many times. The architecture has multiple spatio-temporal convolutional blocks, each with 1 spatial graph convolution layer and 2 temporal gated convolution layers, to improve learning efficiency. These blocks are designed with bottleneck strategies and residual connections. Lastly, the final output is generated by the output fully connected layer. Every image is indicated as a node in a graph, which is how the graph is created. Features extracted from the images serve as node attributes. The edges are defined based on spatial relationships between nodes, which represent similarities between different lesion images based on their features. It is given in Eq. (7)

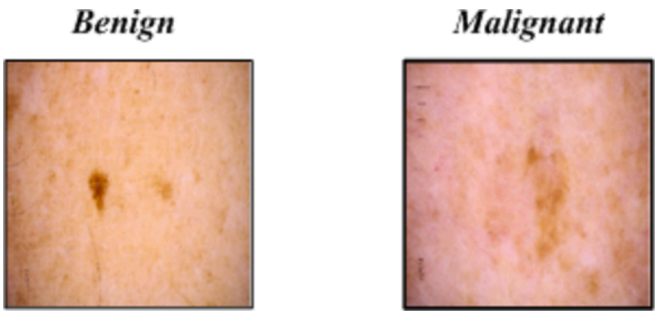


Fig. 3. Classified Result of ISIC 2020 Challenge dataset.

$$Y = \varphi(ZWV + a) \tag{7}$$

here, W indicates the Weight matrix for the graph convolution operation, Z indicates the Input feature matrix for the graph, Y indicates the output after applying the activation function, V is the Matrix of edge weights, a represents the bias term and φ is the activation function. Time-Spatial Graph Multiple spatiotemporal convolutional blocks are used in the convolutional layer. Convolution of Spatial Graphs Eq. (8) provides the spatial dependencies among the lesion features.

$$Y_s^{oce} = \varphi\left(\widetilde{Z}^{(j)} W_{s-j} W_1^{oce} + \widetilde{Z}^{(0)} W_2^{oce} + a^{oce}\right) \tag{8}$$

here, $\widetilde{Z}^j$ is the standardised feature matrix, φ is the activation function, W_1^{oce} indicates the Weight matrix specifically for the original spatial convolutional operation, W_2^{oce} denotes weight matrix for further spatial feature learning,Z is denoted as the Input feature matrix for the graph, Y_s^{oce} is represented as the Output from the spatial graph convolution, a^{zcs} are the learnable parameters, and j is the kernel size. The dilated convolutions within the graph layers are implemented to capture long-range dependencies in both spatial and temporal dimensions. This allows learning complex patterns related to melanoma progression over time. It is given in Eq. (9).

$$A_s^{zcs} = \sum_{j=0}^{j=1} \varphi\left(\widetilde{K}_{s-j:s} W_{s-j} V_j^{zcs} + a^{zcs}\right) \tag{9}$$

here, $\widetilde{K}_{s-j:s}$ are the dynamic Spatio-temporal correlations, s is the time, $V_{j,1}^{Zcs}$ are the learnable parameters, W is the weight matrix for the graph convolution operation, a^{zcs} are the learnable parameters and A_s^{zcs} is the Spatio-temporal correlation. A multi-range attention mechanism is integrated to weigh the significance of various features extracted from different images as it helps in focusing on critical characteristics that differentiate benign from malignant lesions. It is given in Eq. (10).

$$A_s^{zcs} = \sum_{j=0} \varphi\left(\widetilde{k}_{s-j:s} W_{j,1}^{zcs} + \widetilde{K}_{s:s-j} W_{s-j} W_{j,2}^{zcs} + a^{zcs}\right) \tag{10}$$

here, A_s^{zcs} are the Spatio-temporal correlations, W indicates the weight matrix for the graph convolution operation,$W_{j,1}^{zcs}$ *and* $W_{j,2}^{zcs}$ is the parameter, φ is the activation function and a^{zcs} are the learnable parameters. After processing through multiple convolutional layers, the output is passed through a fully connected layer designed for detecting the melanoma. It is given in Eq. (11).

$$F = sigmoid\left(V^f\left[A_s^{oce}, A_s^{zcs}\right] + a^f\right) \tag{11}$$

here, s is the time, V^f, a^f are the learnable parameters, and A_s^{oce} A_s^{zcs} are the Spatio-temporal correlations and to manage the relative relevance of two attributes, F defines a completely connected layer's output. The detected melanoma is classified as malignant and benign using the Eq. (12).

$$A_s = F\Theta A_s^{oce} + (1 - F)\Theta A_s^{zcs} \tag{12}$$

here, Θ is the Model parameter, A_s is the final classification output indicating whether a lesion is malignant or benign,F defines an Output from fully connected layer indicating the probability of the lesion being malignant and A_s^{oce} *and* A_s^{zcs} are the dynamics Spatio-temporal correlations. The Classified Result of ISIC 2020 Challenge dataset is displayed in Fig. 3.

The STJGCN has successfully detected the melanoma and classified it as benign and malignant from the pre-processed images of the ISIC 2020 Challenge Dataset. The RPOA is utilized to enhance the weight factor of STJGCN.

3.4. Optimization using red panda Optimization Algorithm (RPOA)

In this Section, RPOA is discussed [24]. The parameter A_s *and* F of STJGCN is optimized using RPOA. The STJGCN parameters are optimized to increase the accuracy and reduce the loss leading to efficient detection and classification of melanoma. A novel bio-inspired metaheuristic algorithm that mimics the actions of red pandas in the wild is called the Red Panda Optimization Algorithm (RPOA). Red pandas forage and climb trees to rest, two of their primary natural habits, which served as the inspiration for the primary design concept of RPOA. The two stages of the proposed RPOA approach are represented mathematically as exploration, which mimics the red panda's foraging behavior, and exploitation, the movement is similar to that of red pandas climbing trees. The red panda's foraging approach, which is according to its superior sight, hearing, and smell abilities, as well as its exceptional tree-climbing skills, is one of its most remarkable natural behaviors. The design of the proposed RPOA technique, which is described below, is based on mathematical Modelling of these red panda natural behaviors.

Step 1: Initialization

The population-based metaheuristic algorithm, which is the proposed RPOA technique, includes red pandas as members. In an RPOA design, every red panda represents a possible fix for the problem, and depending on where it is in the search spa, it recommends certain solutions for the problem variables. The population of the red panda is given in Eq. (13)

$$w_{h,i} = ka_h + q_{h,i} \cdot (ta_i - ka_i), \quad h = 1, 2, \ldots\ldots, M, \; i = 1, 2, \ldots\ldots, l \tag{13}$$

here,$w_{h,i}$ is indicated as the dimension of the red panda, M is represent as the Number of the red panda, ka_i *and* ta_i is the problem variable's upper and lower bounds, $q_{h,i}$ is the random number in the interval, and l, m are the problem variables.

Step 2: Random generation

The weight bounds are formed arbitrarily generated. Based on a clear hyperparameter condition, the ideal fitness values were chosen.

Step 3: Fitness function

Each red panda's proposed food resource site set, determined by comparing the goal function's values, is modeled. It is shown in Eq. (14).

$$Fitness\,Function = Optimizing\,[A_s \; and \; F] \tag{14}$$

here, A_s denotes the increasing accuracy and F represents the decreasing loss.

Step 4: The Strategy of Red Pandas in Foraging (Exploration phase) for optimizing A_s

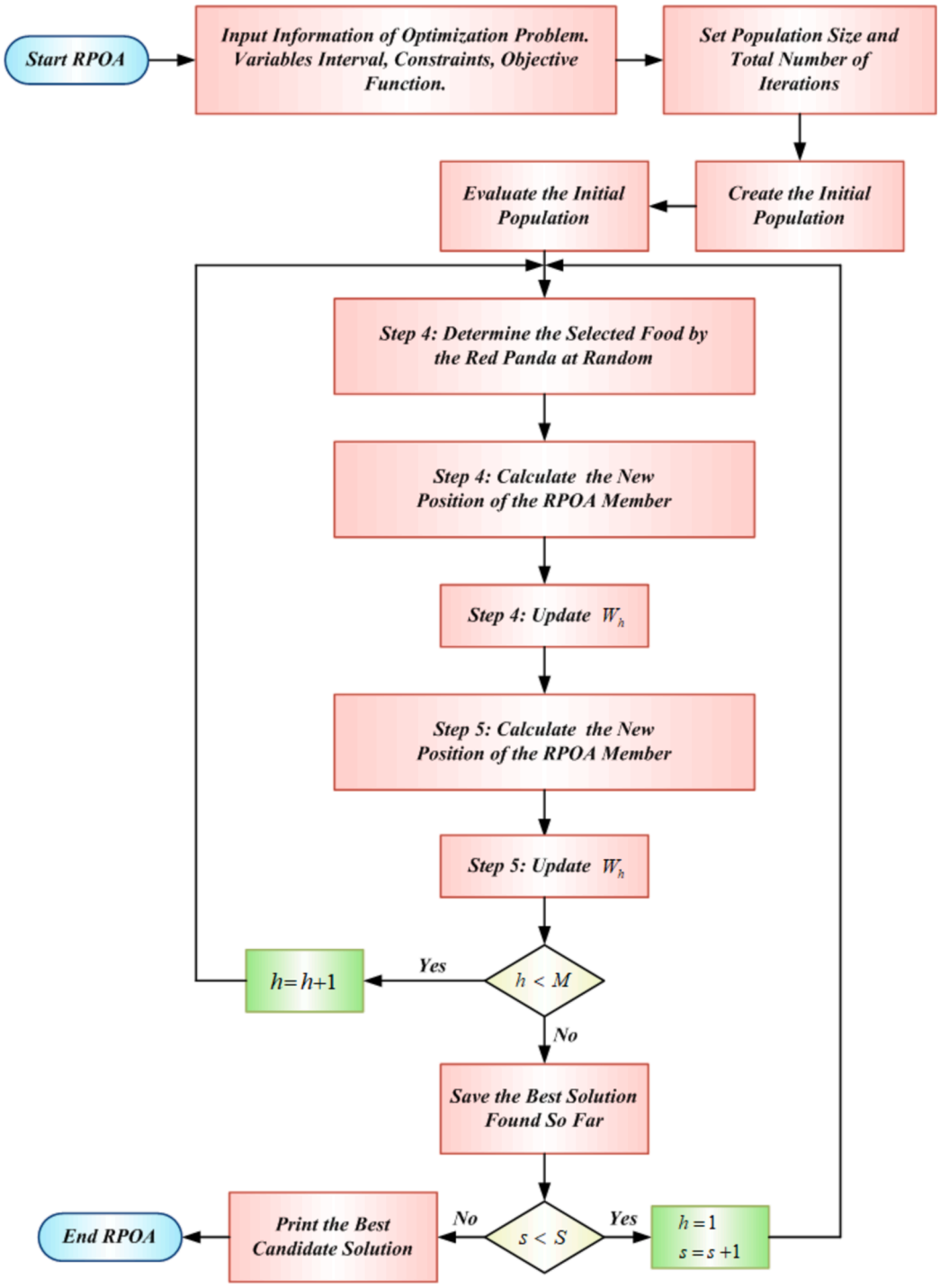


Fig. 4. Flowchart of RPOA.

When approaching the food source, red pandas' positions change dramatically, which improves the proposed algorithm's capacity to investigate and carry out a global search across the problem-solving domain. Based on the movement toward the food source, each red panda's beginning position is established to simulate the behavior of the animals while foraging. It is shown in Eq. (15).

$$W_h^{O1} : w_{h,i}^{O1} = w_{hi} + A_s + q.\left(RER_{hi} - H.w_{h,i}\right) \tag{15}$$

here, $RER_{h,i}$ is indicated as the selected food source, $W_{H,J}^{O1}$ represent as Red Panda's new location and q is the random number in the interval. To imitate the red pandas' natural behavior of climbing trees mechanically, each one's new position is first calculated. It is shown in Eq. (16).

$$W_h = \begin{cases} W_h^{o1} & E_h^{o1} < E_h; \\ W_h & else, \end{cases} \tag{16}$$

here, W_H^{O1} is Red Panda's new location, E_h^{o1} represent as the objective function, W_h is denoted as the *ith* red panda, and E_h is represented as the objective function. This new location is used in place of the linked red panda's previous one if the objective function's value is increased. The flowchart of RPOA is given in Fig. 4.

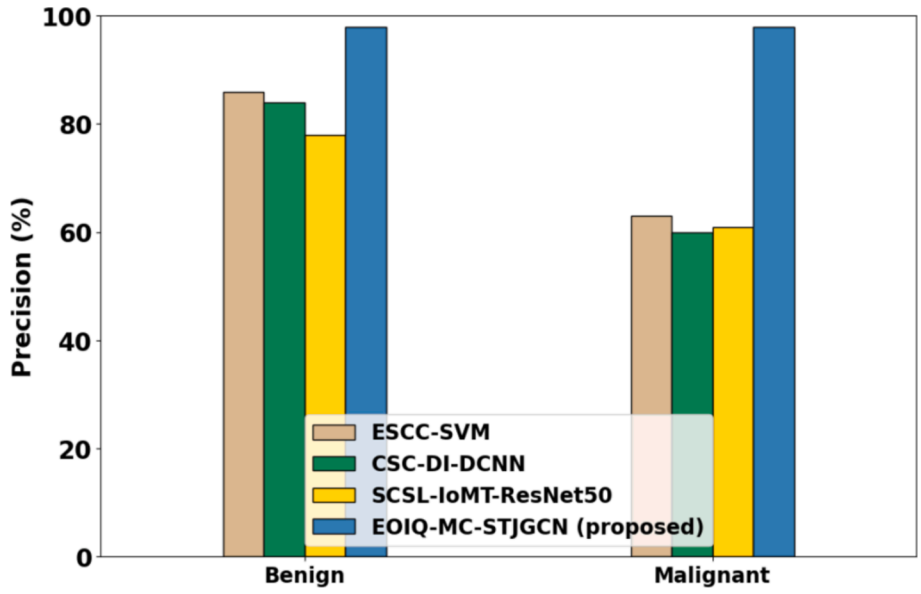


Fig. 5. Performance Analysis of Precision.

Step 5: Skill in Climbing and Resting on the Tree(Exploitation phase) for optimizing F

The first step in determining each panda's new location is to analytically mimic the red pandas' natural climbing tree activity. This new location replaces the previous one used by the linked red panda if the objective function value is increased. Eqn (17) illustrates it.

$$w_{h,i}^{o2} = F + w_{h,i} + \frac{ka_i + q\cdot(ta_i - ka_i)}{s},\ h = 1, 2.....,M, i = 1, 2, ...l,\ s = 1, 2, ..., S \tag{17}$$

here,E_h^{o2} is the objective function,W_H^{O2} is Red Panda's new location, W_h is the *ith* red panda, S is the maximum number of iteration,s denotes the algorithm's iteration counter, and q is the random amount of interval. This new location takes the place of the associated red panda's previous location if the value of the objective function is increased. Eq. (18) displays it.

$$W_h = \begin{cases} W_h^{o2} & E_h^{o2} < E_h \\ W_h & else, \end{cases} \tag{18}$$

here, W_h^{o2} is the red panda's new location and E_h^{o2} indicates the objective function.

Step 6: Termination

RPOA is used to optimize the weight factor value of A_s *and* F from STJGCN. Step 3 will be repeated until it reaches its halting criteria, $h = h+1$ met. Thus, STJGCN is optimized with RPOA for melanoma detection and classification, to improve the detection and classification with greater accuracy and lower loss.

4. Result and discussion

In this paper, Resource management using Effects of Object and Image Quality on Melanoma classification utilizing Spatio Temporal Joint Graph Convolutional Network is discussed.The proposed EQIQ-MC-STJGCN method was implemented in Pythonusing a count of performance measures such as accuracy, precision sensitivity, specificity, F1-Score and RoC. The result of EQIQ-MC-STJGCN approaches was compared with existing ESCC-SVM [14], CSC-DI-DCNN [15] and SCSL-IoMT-ResNet50 [16] techniques.

4.1. Performance measures

This step is essential for whether to explore optimisation algorithms. Performance measures to evaluate to access performance such as accuracy, precision, sensitivity, specificity, F1-Score and RoC.

4.1.1. Accuracy

Eq. (19) provides the formula for computing the accuracy value, this is the proportion of the entire amount of samples to the amount of samples correctly categorized by scheme.

$$accuracy = \frac{TP + TN}{TP + TN + FN + FP} \tag{19}$$

here, TN indicates true negative, TP signifies true positive,FN means false negative and FP indicates false positive.

4.1.2. Precision

Precision computes the count of true positives divided through true positives plus count, false positive number and it is given by the Eq. (20),

$$precision = \frac{TN}{FP + TN} \tag{20}$$

4.1.3. Sensitivity

The percentage of accurately classified samples to all correctly classified occurrences is known as the Sensitivity (recall). Calculating recall allows one to determine the percentage of genuine positives across all positive cases, including false negatives—cases that the model missed. It is especially helpful in situations like medical diagnostics where finding as many positive examples as possible is crucial. It was calculated using Eq. (21)

$$sensitivity = \frac{TN}{TN + FN} \tag{21}$$

4.1.4. Specificity

Specificity estimates proportions of negative, given in Eq. (22),

$$specificity = \frac{TP}{FN + TP} \tag{22}$$

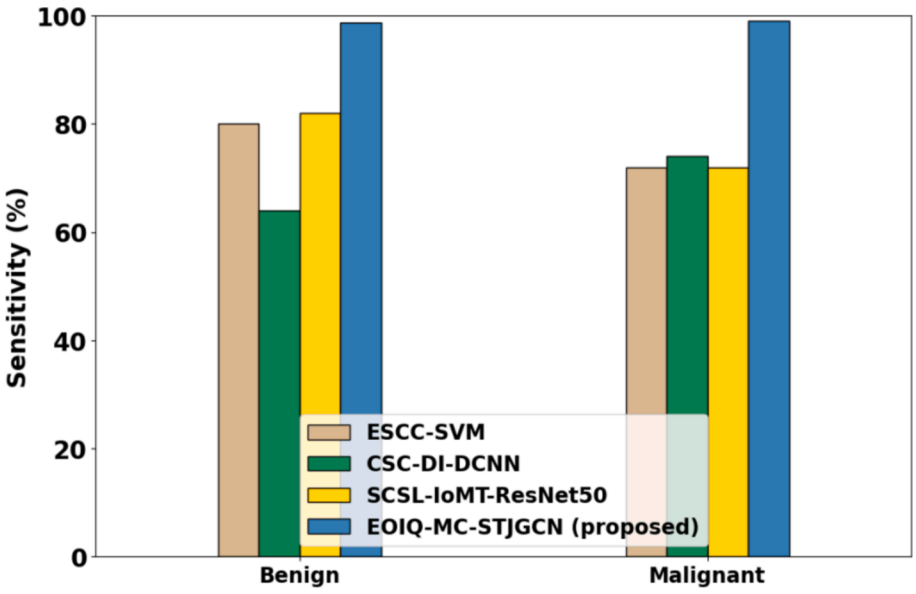


Fig. 6. Performance Analysis of Sensitivity.

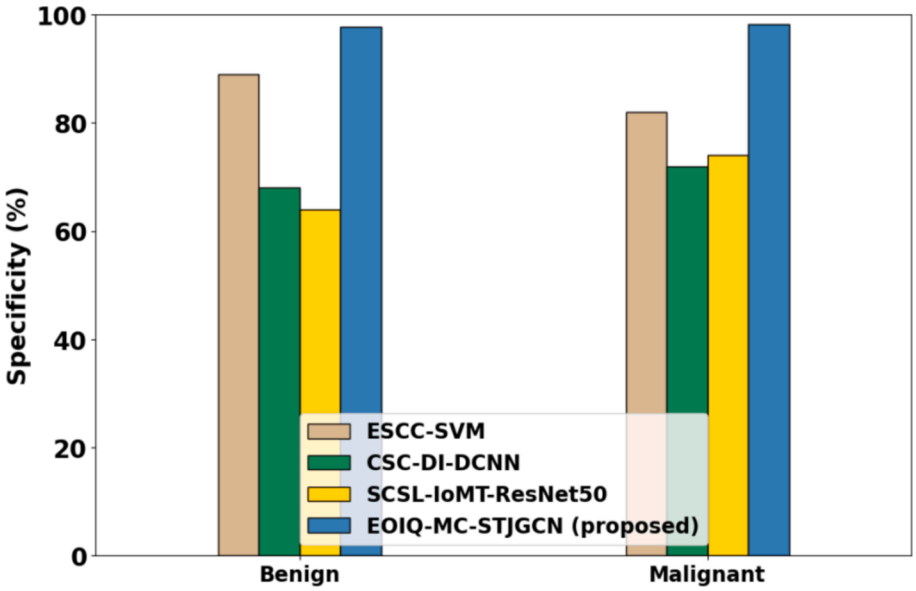


Fig. 7. Performance Analysis of Specificity.

4.1.5. F1-Score

The F-Score, also called as the F1 score, evaluates performance by taking into account both recall and precision. It provides a fair assessment of efficacy by taking the harmonic mean of these two measurements. An elevated F-Score is a result of higher precision and recall levels. The Eq. (23) that follows can be used to get the F-Score.

$$F1 - Score = \frac{2*precision*recall}{precision + recall} \tag{23}$$

4.2. Performance analysis

Figs. 5–12 displays the simulation results of EOIQ-MC-STJGCN method proposed. Proposed EOIQ-MC-STJGCN method is contrasted with existing like ESCC-SVM, CSC-DI-DCNN and SCSL-IoMT-ResNet50respectively.

The performance analysis of precision is illustrated in Fig. 5. The bar graph shows the precision analysis for various models on the ISIC 2020 challenge dataset, namely the categorization of patients with melanoma as either benign or malignant. For benign lesions, the EOIQ-MC-STJGCN (proposed) model exhibits the highest precision, reaching approximately 28.41 %, 32.12 %, and 35.12 % higher than the other models such as ESCC-SVM, CSC-DI-DCNN, and SCSL-IoMT-ResNet50. For malignant melanoma cases, the EOIQ-MC-STJGCN model maintains 31.42 %, 34.02 %, and 31.02 % higher precision than the other models such as ESCC-SVM, CSC-DI-DCNN and SCSL-IoMT-ResNet50 respectively. The EOIQ-MC-STJGCN model consistently outperforms alternative approaches demonstrating its high efficacy in accurately classifying melanoma within the ISIC 2020 dataset.

The performance analysis of Sensitivity is depicted in Fig. 6. The bar graph illustrates the sensitivity performance of various models on the ISIC 2020 challenge dataset, specifically assessing their capability to correctly identify true positive cases for both benign and malignant melanoma classifications. For benign melanoma cases, the EOIQ-MC-STJGCN (proposed) model achieves the highest sensitivity, reaching

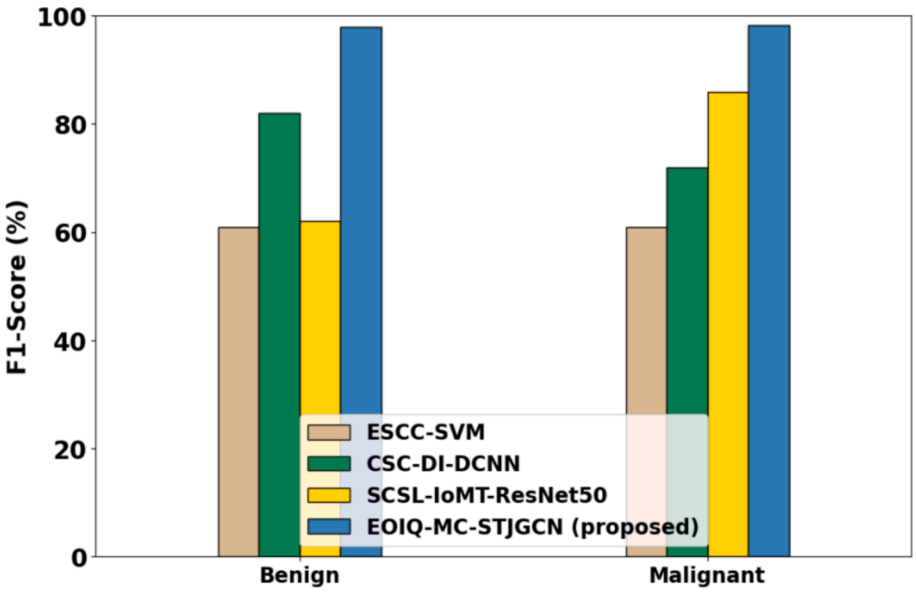


Fig. 8. Performance Analysis of F1-Score.

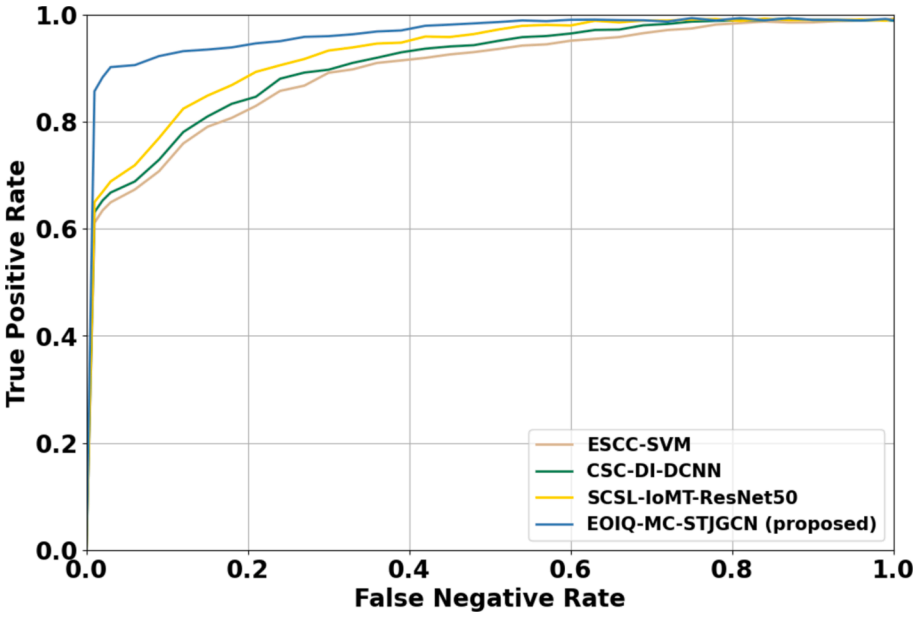


Fig. 9. Performance Analysis of RoC.

17.97 %, 25.68 % and 30.18 % higher than the existing models. For malignant melanoma cases, the EOIQ-MC-STJGCN model again exhibits superior performance, with 25.78 %, 18.51 % and 30.77 % higher sensitivity than the existing models such as ESCC-SVM, CSC-DI-DCNN and SCSL-IoMT-ResNet50 respectively. The EOIQ-MC-STJGCN model demonstrates its high accuracy and effectiveness in identifying true positive cases, which is crucial for accurate melanoma classification in the ISIC 2020 dataset.

The performance analysis of Specificity is depicted in Fig. 7. The bar graph provides a comparative analysis of specificity across various models on the ISIC 2020 dataset, concentrating on their ability to accurately distinguish between malignant and benign lesions and identify true negative cases of melanoma. The proposed EOIQ-MC-STJGCN approach attains 26.46 %, 21.68 % and 30.79 % higher specificity for Benign and 23.79 %, 30.51 % and 20.81 % higher specificity for Malignant when compared with existing approaches such as ESCC-SVM, CSC-DI-DCNN and SCSL-IoMT-ResNet50. This analysis underscores the EOIQ-MC-STJGCN model is a highly reliable approach for minimizing false positives and improving melanoma diagnosis accuracy within the ISIC 2020 dataset.

The performance analysis of the F1-Score is displayed in Fig. 8. The F1-Score for several models applied to the ISIC 2020 dataset is compared in the bar graph, which evaluates how well each model balances recall and precision for the classification of malignant and benign. The proposed EOIQ-MC-STJGCN model achieves the highest F1-Score, approximately 27.46 %, 29.68 % and 32.79 % for benign and 27.62 %, 27.32 % and 35.17 % for malignant cases compared with such as ESCC-SVM, CSC-DI-DCNN and SCSL-IoMT-ResNet50, demonstrating superior capability in maintaining a balanced classification performance.

The performance analysis of RoC is displayed in Fig. 9. The ISIC 2020 challenge dataset's ROC curve study for melanoma classification demonstrates how well different models distinguish between benign and

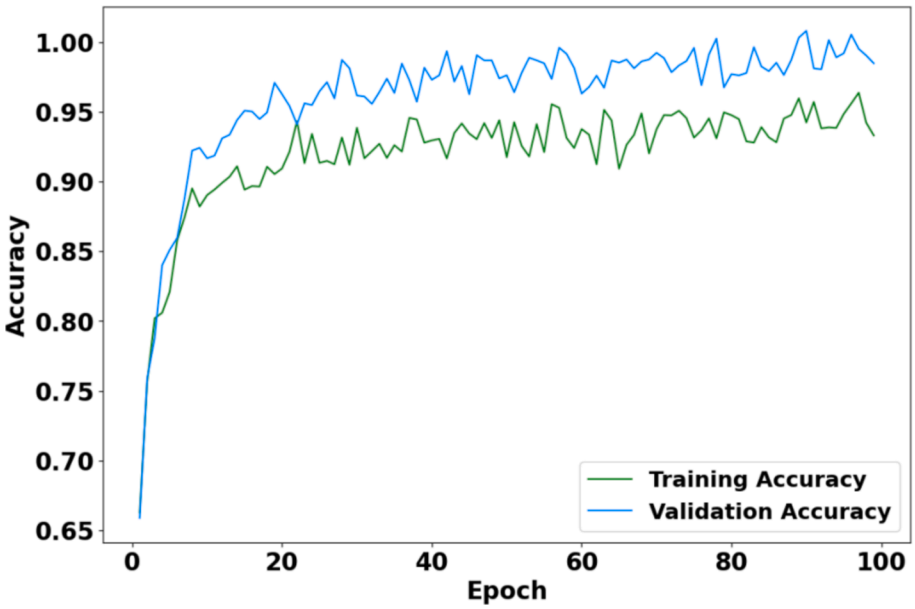


Fig. 10. Accuracy graph for training and validation of EOIQ-MC-STJGCN.

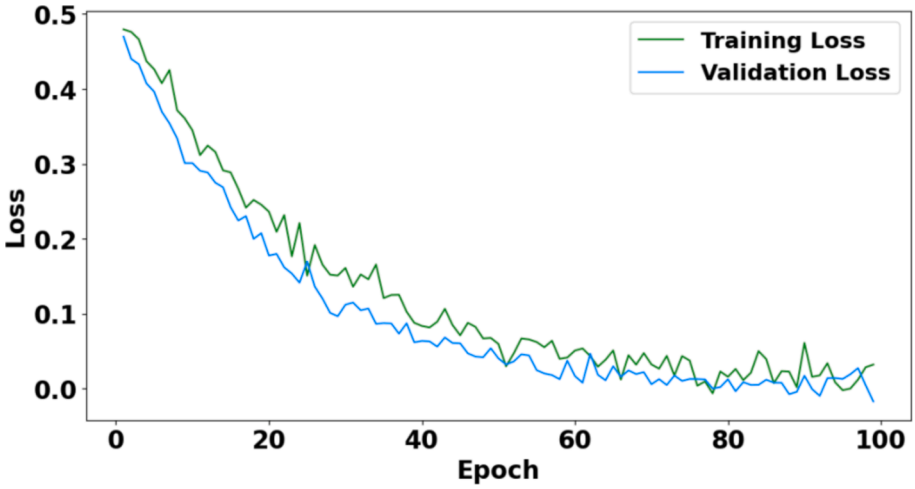


Fig. 11. Loss graphs for training and validation of EOIQ-MC-STJGCN.

malignant skin lesions. With a low false positive rate (1-specificity) and a high true positive rate (sensitivity), the proposed EOIQ-MC-STJGCN model performs better than the competition. The proposed EOIQ-MC-STJGCN model attains a higher value of 0.97 RoC. This analysis underscores the EOIQ-MC-STJGCN model's superior capability in accurately classifying skin lesions, thereby offering a promising tool for clinical applications in melanoma detection.

The Accuracy graph for training and validation of EOIQ-MC-STJGCN is illustrated in Fig. 10. The graph displays the accuracy curves for training and validation for the EOIQ-MC-STJGCN model, which is intended to classify melanoma on the dataset for the ISIC 2020 challenge by differentiating between benign cases and those that are malignant. The blue line, representing training accuracy, exhibits a rapid increase during the initial epochs and peaks at around 0.995. The validation accuracy, shown by the green line, also rises initially but stabilizes around 0.95 after a certain point. The final training accuracy of approximately 0.995 and validation accuracy of 0.95 highlight the importance of further tuning to enhance robustness for classifying benign and malignant melanoma cases accurately.

The Loss graphs for training and validation of EOIQ-MC-STJGCN is illustrated in Fig. 11 for melanoma classification on the ISIC 2020 challenge dataset, which differentiates between benign and malignant cases. The green line represents the training loss, which steadily decreases from approximately 0.49 to around 0.01. The blue line represents the validation loss, initially following a downward trend and reaching a minimum of about 0.03. However, after around 60 epochs, the validation loss starts to fluctuate and even slightly increases, thereby reducing the capability to generalize effectively to new data. The EOIQ-MC-STJGCN model achieves a final training loss of approximately 0.01 and a validation loss of around 0.03 in melanoma classification.

The Confusion Matrix of the proposed approach on the ISIC 2020 challenge dataset is displayed in Fig. 12. Using the ISIC 2020 dermoscopy image dataset, the confusion matrix demonstrates the excellent performance of the proposed EOIQ-MC-STJGCN model for melanoma detection and classification. The model achieved 5,487 true negatives (benign correctly classified as benign) and 5,490 true positives (malignant correctly classified as malignant), with only 4 false positives (benign misclassified as malignant) and 1 false negative (malignant misclassified as benign). This reflects a very high accuracy, with both precision and recall for malignant cases being close to perfect. According

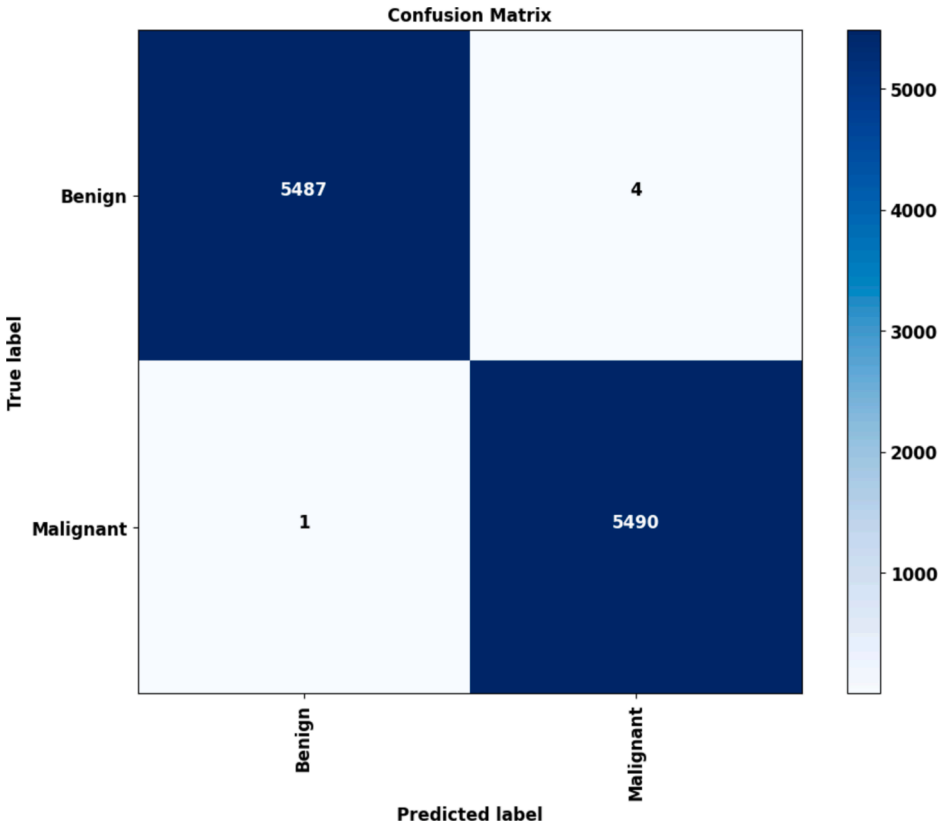


Fig. 12. Confusion Matrix of the proposed approach on the ISIC 2020 challenge dataset.

Table 2
Outcome of proposed model with different architecture.

Parameter	Setting 1	Setting 2	Setting 3
Learning Rate	0.001	0.005	0.007
Dropout Rate	0.3	0.4	0.5
Number of Layers	3	4	5
Nodes per Layer	64	128	256
Activation Function	ReLU	Leaky ReLU	Sigmoid
Batch Size	32	64	128
Epoch	100	150	200
Optimizer	RPOA	Adam	Stochastic Gradient Descent (SGD)
Accuracy (%)	97.86	94.14	92.37

to these findings, the proposed EOIQ-MC-STJGCN model is a very promising tool for the detection and classification of melanoma because it can effectively differentiate between benign and malignant lesions.

The Table 2 illustrates the outcomes of the proposed model with various architectural settings for melanoma classification on the ISIC 2020 dataset. Setting 1, the proposed and most effective configuration, achieved the highest accuracy of 97.86 %. ReLU activation, three layers with 64 nodes each, a batch size of 32, 100 epochs, a learning rate of 0.001, a dropout rate of 0.3, and the RPOA optimizer were all used. In contrast, Setting 2 featured a learning rate of 0.005, dropout rate of 0.4, 4 layers with 128 nodes per layer, Leaky ReLU activation, batch size of 64, 150 epochs, and the Adam optimizer, resulting in a lower accuracy of 94.14 %. Setting 3 used a learning rate of 0.007, dropout rate of 0.5, and 5 layers with 256 nodes per layer, sigmoid activation, batch size of 128, 200 epochs, and Stochastic Gradient Descent (SGD) optimizer, yielding the lowest accuracy at 92.37 %. This analysis underscores Setting 1 as the optimal configuration, combining efficiency and high performance.

The Table 3 presents a comparative analyses of the proposed EOIQ-MC-STJGCN method with existing approaches for melanoma classification. The proposed EOIQ-MC-STJGCN method significantly outperforms others, achieving the highest accuracy of 97.86 %, precision of 97.67 %, sensitivity of 98.6 %, specificity of 97.65 %, F1-score of 97.86 %, and RoC of 0.97. In comparison, other methods show lower performance, such as CSC-DI-DCNN with an accuracy of 84.34 % and RoC of 0.85, and SCD-DI-EfficientNetV2-M with 81.64 % accuracy and RoC of 0.84. The SCSL-IoMT-ResNet50 achieves 81.34 % accuracy and a lower F1-score of 72.64 %. The MRF-DI-SVM method performs with 79.34 % accuracy and a RoC of 0.75, while SLC-RDCNN records 80.21 %

Table 3
Performance Analyses of Proposed with Existing Approaches.

Methods	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	RoC
ESCC-SVM [14]	78.32	79.32	84.1	71.74	75.2	0.79
CSC-DI-DCNN [15]	84.34	81.31	69.43	69.64	74.32	0.85
SCSL-IoMT-ResNet50 [16]	81.34	74.65	72.54	75.31	72.64	0.78
SCD-CNN [17]	76.38	71.25	75.3	73.61	70.21	0.80
SCD-DI-EfficientNetV2-M [18]	81.64	80.41	69.4	69.31	69.32	0.84
MRF-DI-SVM [19]	79.34	71.63	72.63	72.31	72.45	0.75
SLC-RDCNN [20]	80.21	70.31	68.71	70.41	73.14	0.71
EOIQ-MC-STJGCN (Proposed)	97.86	97.67	98.6	97.65	97.86	0.97

Table 4
Comparison of Proposed EOIQ-MC-STJGCN with various methods.

Methods	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	RoC
DSA-Conv-LSTM [11]	78.32	79.32	84.1	71.74	75.2	0.79
AFSTGCN [12]	84.34	81.31	69.43	69.64	74.32	0.85
STFormer [13]	81.34	74.65	72.54	75.31	72.64	0.78
EOIQ-MC-STJGCN (Proposed)	97.86	97.67	98.6	97.65	97.86	0.97

accuracy and 0.71 RoC. The consistently higher metrics of the EOIQ-MC-STJGCN across all metrics establish it as the most effective model for melanoma classification, particularly for its robustness and precision in detecting melanoma from the ISIC 2020 challenging dataset.

The Table 4 presents a comparison ofProposed EOIQ-MC-STJGCN with various methods used to classify melanoma from the ISIC 2020 challenging dataset. The proposed EOIQ-MC-STJGCN method outperforms all other methods in all evaluation metrics. It attains a 97.86 % accuracy, a 97.67 % precision, a 98.6 % sensitivity, a 97.65 % specificity, a 97.86 % F1-score, and a ROC of 0.97. This represents that the proposed method is highly accurate and reliable in classifying melanoma from the dataset, making it an efficient and effective solution.

4.3. Discussion

The proposed EOIQ-MC-STJGCN approach demonstrates substantial improvements in melanoma classification, better than existing methods, attaining 97.86 % accuracy on the ISIC 2020 Challenge Dataset. The preprocessing through the Information Exchange Multi-Bernoulli Filter (IEMBF) effectively removes noise and extraneous elements like hair and rulers, ensuring higher image quality. By incorporating the Red Panda Optimization Algorithm (RPOA) to fine-tune the Spatio Temporal Joint Graph Convolutional Network (STJGCN) parameters, the model more effectively captures features from dermoscopy images, addressing limitations in previous approaches. Specifically, EOIQ-MC-STJGCN delivers 27.46 %, 29.68 % and 32.79 % higher F1-Score, 23.79 %, 30.51 % and 20.81 % higher specificity and 17.97 %, 25.68 % and 30.18 % higher sensitivity than ESCC-SVM, CSC-DI-DCNN and SCSL-IoMT-ResNet50 respectively. In clinical practice, EOIQ-MC-STJGCN could reduce diagnostic errors and enable earlier melanoma detection, particularly in resource-limited or time-sensitive environments where immediate and accurate diagnostic tools are essential. The model's high sensitivity and specificity mean it could serve as a reliable CAD (Computer-Aided Diagnosis) tool to assist dermatologists in distinguishing between benign and malignant lesions, reducing unnecessary biopsies and ensuring that patients with malignant lesions receive timely intervention.

5. Conclusion

In this section, the detection and classification of melanoma using the proposed EOIQ-MC-STJGCN approach based on the ISIC 2020 Challenge dataset was successfully implemented in Python. The Information Exchange Multi-Bernoulli-Filter (IEMBF) effectively reduces noise and enhances image quality, which is critical for accurate melanoma diagnosis. The integration of the Red Panda Optimization Algorithm (RPOA) further improves the performance of the STJGCN, resulting in a significant improvement in classification accuracy. The outcomes reveal that the proposed EOIQ-MC-STJGCN model attains 23.79 %, 30.51 % and 20.81 % higher specificity and 31.42 %, 34.02 % and 31.02 % higher precision outperforming existing approaches such as ESCC-SVM, CSC-DI-DCNN and SCSL-IoMT-ResNet50. The increased model complexity may impact interpretability, making it more challenging to implement in clinical settings. Additionally, the computational demands associated with training and deploying the model could limit its accessibility in resource-constrained environments. Future research directions should focus on exploring lighter network structures that maintain high performance while reducing computational costs and improving interpretability. Furthermore, efforts should be made to enhance the generalization ability of the model across diverse datasets and real-world clinical scenarios, ensuring robust performance in various imaging conditions and among different populations.

6. Data availability Statement

The lack of newly created or inspected data makes this study exempt from the data sharing policy.

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CRediT authorship contribution statement

V. Suryanarayana: Supervision. **B. Prabhu Shankar:** . **Rama Devi Burri:** Supervision. **T. Priyanka:** Supervision. **Ravi Kumar Saidala:** Writing – original draft. **A. Sasi Kumar:** . **Piyush Chauhan:** Supervision. **Jagdish Chandra Patni:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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